C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 3.01

Indices, Surds & Polynomials

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

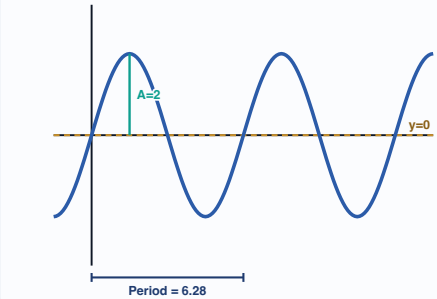
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	**Q2** C	**Q3** B	**Q4** A	**Q5** A
Q6 A	**Q7** A	**Q8** A	**Q9** A	**Q10** C
Q11 D	**Q12** B	**Q13** D	**Q14** A	**Q15** C
Q16 B	**Q17** A	**Q18** B	**Q19** B	**Q20** A
Q21 A	**Q22** A	**Q23** B	**Q24** D	**Q25** B

Indices, Surds & Polynomials Practice Exercises

1 MCQ

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.



- **A** 57 units per second ✓
- **B** 19 units per second
- **C** $\frac{19}{3}$ units per second
- **D** 21 units per second

WORKED SOLUTION & VERIFICATION

To find the rate of change of y with respect to time, apply the chain rule for connected rates of change:

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

1. Differentiate y with respect to x:

$$\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$$

2. Evaluate $\frac{dy}{dx}$ at $x = 2$:

$$\left.\frac{dy}{dx}\right|_{x=2} = 6(2)^2 - 5 = 6(4) - 5 = 24 - 5 = 19$$

3. Multiply by $\frac{dx}{dt} = 3$:

$$\frac{dy}{dt} = 19 \times 3 = 57 \text{ units per second}$$

Common errors:
- 19 corresponds to finding $\frac{dy}{dx}$ only and neglecting $\frac{dx}{dt}$.
- $\frac{19}{3}$ corresponds to inverting the chain rule (dividing instead of multiplying by $\frac{dx}{dt}$).
- 21 results from incorrectly differentiating $2x^3$ as $6x$ instead of $6x^2$. [M1][A1]

2 MCQ

Consider the cubic polynomial $f(x) = x^3 - 5x^2 + 10x - 20$. Find the remainder $f(2)$ upon division by $(x - 2)$.

A -10

B -7

C **-12** ✓

D -22

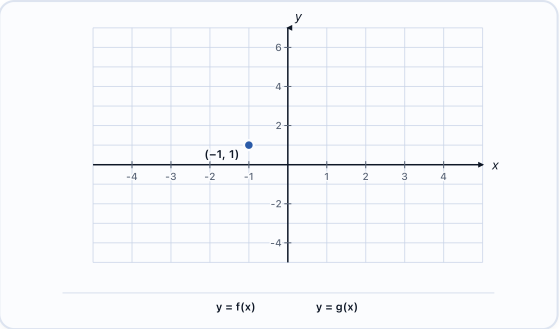
WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** By the Remainder Theorem, $f(2) = 2^3 - 5(2^2) + 10(2) - 20 = -12$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: -12. **[A1]**

3 MCQ

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$.
Find the area of the region enclosed between the curve and the x-axis.



A 27

B 36 ✓

C 54

D 72

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** To find the area enclosed between the curve and the x-axis, integrate y with respect to x from $x = 0$ to $x = 6$:

$$\text{Area} = \int_0^6 (6x - x^2)\,dx$$

Integrate term by term:

$$\int (6x - x^2)\,dx = \left[\frac{6x^2}{2} - \frac{x^3}{3}\right]_0^6 = \left[3x^2 - \frac{x^3}{3}\right]_0^6$$

Substitute the limits:

$$\left(3(6)^2 - \frac{6^3}{3}\right) - 0 = 3(36) - \frac{216}{3} = 108 - 72 = 36$$

Common errors:

- Estimating the region as a triangle with base 6 and maximum height 9 yields $\frac{1}{2} \times 6 \times 9 = 27$.
- Taking the bounding rectangle gives $6 \times 9 = 54$.
- Integrating $6x$ as $6x^2$ without dividing by 2 gives $216 - 72 = 144$, or subtracting $108 - 36 = 72$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 36. **[A1]**

4 MCQ

Solve the logarithmic equation $\log_2(5x+6) - \log_2(x) = 3$.

- **A** $x = 2$ ✓
- **B** $x = 6$
- **C** $x = \frac{1}{2}$
- **D** $x = -3$

WORKED SOLUTION & VERIFICATION

1. Apply the logarithm quotient law $\log_a(A) - \log_a(B) = \log_a(\frac{A}{B})$:

$$\log_2(\frac{5x+6}{x}) = 3$$

2. Convert the equation from logarithmic to exponential form:

$$\frac{5x+6}{x} = 2^3 = 8$$

3. Multiply through by x (noting $x > 0$):

$$5x + 6 = 8x \implies 3x = 6 \implies x = 2$$

4. Checking the domain:

Both $5(2) + 6 = 16 > 0$ and $2 > 0$, so $x = 2$ is a valid solution.

Common misconceptions:

- Calculating $2^3 = 6$ instead of 8 yields $5x + 6 = 6x \implies x = 6$.
- Incorrectly subtracting the arguments directly as $\log_2((5x+6) - x) = 3$ leads to $4x + 6 = 8 \implies x = \frac{1}{2}$.
- Forgetting to exponentiate the base and equating $\frac{5x+6}{x} = 3$ yields $x = -3$, which also violates the logarithm domain. [M1][A1]

5 MCQ

Find the value of the integer $n \geq 3$ that satisfies the equation

$$^{n}\mathrm{P}_{3} = 10 \times {}^{n}\mathrm{C}_{2}$$

A $n = 7$ ✓

B $n = 12$

C $n = 5$

D $n = 8$

WORKED SOLUTION & VERIFICATION

Using the algebraic definitions of permutations and combinations:

$$^{n}\mathrm{P}_{3} = \frac{n!}{(n-3)!} = n(n-1)(n-2)$$

$$^{n}\mathrm{C}_{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2}$$

Substitute these into the given equation:

$$n(n-1)(n-2) = 10 \times \frac{n(n-1)}{2}$$

$$n(n-1)(n-2) = 5n(n-1)$$

Since $n \geq 3$, we have $n(n-1) \neq 0$. Dividing both sides by $n(n-1)$ gives:

$$n - 2 = 5 \implies n = 7$$

Common misconceptions:

- Forgetting the $2!$ denominator in $^{n}\mathrm{C}_{2}$ (treating it as $^{n}\mathrm{P}_{2}$) gives $n - 2 = 10 \implies n = 12$.
- Dividing by $n(n-1)$ and erroneously writing $n = 5$ instead of $n - 2 = 5$ gives $n = 5$.
- Confusing the factorial simplification to set $n - 3 = 5$ leads to $n = 8$. [M1][A1]

6 MCQ

Find the remainder when the polynomial $f(x) = 4x^3 - 6x^2 + 3x - 1$ is divided by $2x - 1$.

A $-\frac{1}{2}$ ✓	**B** $\frac{1}{2}$
C $-\frac{9}{2}$	**D** 0

WORKED SOLUTION & VERIFICATION

By the Remainder Theorem, when a polynomial $f(x)$ is divided by a linear factor of the form $ax - b$, the remainder is $f(\frac{b}{a})$.

Here, the divisor is $2x - 1$, so we set $2x - 1 = 0 \implies x = \frac{1}{2}$.

Substitute $x = \frac{1}{2}$ into $f(x)$:

$$f(\frac{1}{2}) = 4(\frac{1}{2})^3 - 6(\frac{1}{2})^2 + 3(\frac{1}{2}) - 1$$

$$= 4(\frac{1}{8}) - 6(\frac{1}{4}) + \frac{3}{2} - 1$$

$$= \frac{1}{2} - \frac{3}{2} + \frac{3}{2} - 1 = -\frac{1}{2}$$

Therefore, the remainder is $-\frac{1}{2}$.

Common misconceptions:

- Solving $2x - 1 = 0$ incorrectly as $x = -\frac{1}{2}$ gives $f(-\frac{1}{2}) = -\frac{9}{2}$.
- Arithmetic sign errors when simplifying the terms often lead to $\frac{1}{2}$ or 0. [M1][A1]

7 MCQ

In $\triangle ABC$, sides of lengths 32 and 10 meet at an included angle of 60° opposite side x. Find the squared length $f = x^2$ by the Law of Cosines.

A **804** ✓	**B** 803
C 820	**D** 809

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** By the Law of Cosines, $x^2 = 32^2 + 10^2 - 2(32)(10)\cos(60^\circ) = 804$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 804. **[A1]**

8 MCQ

The position vectors of points A and B relative to an origin O are $\vec{OA} = 2\mathbf{i} - 3\mathbf{j}$ and $\vec{OB} = 7\mathbf{i} + 9\mathbf{j}$. Find the unit vector in the direction of $\vec{AB}$.

A $\frac{1}{13}(5\mathbf{i} + 12\mathbf{j})$ ✓

B $\frac{1}{13}(-5\mathbf{i} - 12\mathbf{j})$

C $\frac{1}{\sqrt{61}}(5\mathbf{i} + 6\mathbf{j})$

D $\frac{1}{15}(9\mathbf{i} + 12\mathbf{j})$

WORKED SOLUTION & VERIFICATION

First, find the displacement vector $\vec{AB}$:

$$\vec{AB} = \vec{OB} - \vec{OA} = (7\mathbf{i} + 9\mathbf{j}) - (2\mathbf{i} - 3\mathbf{j}) = (7 - 2)\mathbf{i} + (9 - (-3))\mathbf{j} = 5\mathbf{i} + 12\mathbf{j}.$$

Next, calculate the magnitude of $\vec{AB}$:

$$|\vec{AB}| = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

The unit vector in the direction of $\vec{AB}$ is:

$$\frac{\vec{AB}}{|\vec{AB}|} = \frac{1}{13}(5\mathbf{i} + 12\mathbf{j}).$$

- Choice B gives the unit vector for $\vec{BA} = \vec{OA} - \vec{OB}$, which is in the opposite direction.
- Choice C results from an arithmetic sign error in the $\mathbf{j}$-component: $9 - 3 = 6$ instead of $9 - (-3) = 12$.
- Choice D results from incorrectly adding the position vectors $\vec{OA} + \vec{OB}$. [M1][A1]

9 MCQ

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $x - 2$ and leaves a remainder of -24 when divided by $x + 1$, where a and b are constants.
Find the remainder when $P(x)$ is divided by $x - 3$.

A 12 ✓

B -150

C -30

D 60

WORKED SOLUTION & VERIFICATION

Step 1: Use the Factor Theorem for the factor $x - 2$.
Since $x - 2$ is a factor, $P(2) = 0$:

$$P(2) = 2(2)^3 + a(2)^2 + b(2) - 6 = 0$$

$$16 + 4a + 2b - 6 = 0 \implies 4a + 2b = -10 \implies 2a + b = -5 \quad \text{— (1)}$$

Step 2: Use the Remainder Theorem for division by $x + 1$.
Since the remainder is -24, $P(-1) = -24$:

$$P(-1) = 2(-1)^3 + a(-1)^2 + b(-1) - 6 = -24$$

$$-2 + a - b - 6 = -24 \implies a - b - 8 = -24 \implies a - b = -16 \quad \text{— (2)}$$

Step 3: Solve the simultaneous equations.
Adding (1) and (2):

$$(2a + b) + (a - b) = -5 + (-16)$$

$$3a = -21 \implies a = -7$$

Substitute $a = -7$ into (2):

$$-7 - b = -16 \implies b = 9$$

Thus, the polynomial is:

$$P(x) = 2x^3 - 7x^2 + 9x - 6$$

Step 4: Find the remainder when $P(x)$ is divided by $x - 3$.
By the Remainder Theorem, the remainder is $P(3)$:

$$P(3) = 2(3)^3 - 7(3)^2 + 9(3) - 6 = 2(27) - 7(9) + 27 - 6 = 54 - 63 + 27 - 6 = 12$$

Common Pitfalls:
- Evaluating $P(-3)$ instead of $P(3)$ yields -150.
- Using incorrect signs when substituting into the Factor/Remainder Theorems (e.g., testing $P(-2) = 0$ and $P(1) = -24$) leads to -30.
- A sign error setting the remainder to $+24$ leads to 60. [M1][A1]

10 MCQ

In the expansion of $(1+ax)^n$ in ascending powers of x, the coefficient of x^2 is 60 and the coefficient of x^3 is 160, where n is a positive integer and $a > 0$.
A cubic polynomial $P(x)$, in which the coefficient of x^3 is 1, has $(x-a)^2$ as a factor and leaves a remainder of n when divided by $(x-1)$.
Find the remainder when $P(x)$ is divided by $(x+a)$.

A -48

B -112

C 48 ✓

D 0

WORKED SOLUTION & VERIFICATION

Step 1: Determine n and a using the Binomial Theorem.
In the expansion of $(1+ax)^n$:

$$\text{Term in } x^2 = \binom{n}{2}(ax)^2 = \frac{n(n-1)}{2}a^2x^2 = 60x^2 \implies n(n-1)a^2 = 120$$

$$\text{Term in } x^3 = \binom{n}{3}(ax)^3 = \frac{n(n-1)(n-2)}{6}a^3x^3 = 160x^3 \implies n(n-1)(n-2)a^3 = 960$$

Dividing the equation for x^3 by the equation for x^2:

$$\frac{n(n-1)(n-2)a^3}{n(n-1)a^2} = \frac{960}{120} \implies (n-2)a = 8 \implies a = \frac{8}{n-2}$$

Substitute $a = \frac{8}{n-2}$ into $n(n-1)a^2 = 120$:

$$n(n-1)\left(\frac{8}{n-2}\right)^2 = 120 \implies \frac{64n(n-1)}{(n-2)^2} = 120$$

Dividing both sides by 8:

$$\frac{8n(n-1)}{(n-2)^2} = 15 \implies 8(n^2-n) = 15(n^2-4n+4)$$

$$8n^2-8n = 15n^2-60n+60 \implies 7n^2-52n+60 = 0$$

Factoring the quadratic:

$$(7n-30)(n-6) = 0$$

Since n must be a positive integer, we have $n = 6$.
Then $a = \frac{8}{6-2} = \frac{8}{4} = 2$.
Step 2: Determine the polynomial $P(x)$.
Since $P(x)$ is a cubic polynomial with leading coefficient 1 and has $(x-a)^2 = (x-2)^2$ as a factor, it can be written as:

$$P(x) = (x-2)^2(x-c)$$

for some constant c.
By the Remainder Theorem, the remainder when $P(x)$ is divided by $(x-1)$ is $P(1)$:

$$P(1) = (1-2)^2(1-c) = 1 \cdot (1-c) = 1-c$$

Given that this remainder is $n = 6$:

$$1 - c = 6 \implies c = -5$$

Thus, $P(x) = (x-2)^2(x+5)$.
Step 3: Evaluate the remainder when $P(x)$ is divided by $(x+a) = (x+2)$.
By the Remainder Theorem, the required remainder is $P(-2)$:

$$P(-2) = (-2-2)^2(-2+5) = (-4)^2 \cdot 3 = 16 \cdot 3 = 48$$

Misconception Analysis:
- -48 occurs from a sign error when evaluating $(-4)^2$ as -16.
- -112 occurs from incorrectly solving $1 - c = 6$ as $c = 5$, leading to $P(-2) = 16(-7) = -112$.
- 0 occurs from confusing the divisor $(x+a)$ with the factor $(x-a)$. [M1][A1]

11 MCQ

When the polynomial $\mathrm{P}(x) = 2x^3 - 3x^2 + ax - 8$ is divided by $x - 3$, the remainder is 10. What is the value of the constant a?

A 3

B -9

C -33

D -3 ✓

WORKED SOLUTION & VERIFICATION

According to the Remainder Theorem, when a polynomial $\mathrm{P}(x)$ is divided by $x - c$, the remainder is equal to $\mathrm{P}(c)$.
Here, the divisor is $x - 3$, so $c = 3$. We evaluate $\mathrm{P}(3)$:

$$\mathrm{P}(3) = 2(3)^3 - 3(3)^2 + a(3) - 8$$

$$\mathrm{P}(3) = 2(27) - 3(9) + 3a - 8 = 54 - 27 + 3a - 8 = 19 + 3a$$

Given that the remainder is 10:

$$19 + 3a = 10$$

$$3a = 10 - 19 = -9$$

$$a = -3$$

Common misconceptions:
- Choosing 3 results from a sign error when rearranging $19 + 3a = 10$ to $3a = 19 - 10 = 9$.
- Choosing -9 results from stopping at $3a = -9$ without dividing by the coefficient of a.
- Choosing -33 results from evaluating $\mathrm{P}(-3)$ instead of $\mathrm{P}(3)$. [M1][A1]

12 MCQ

In the expansion of $(1+ax)^n$ in ascending powers of x, where n is a positive integer and a is a constant, the coefficient of x is -12 and the coefficient of x^2 is 60. Find the coefficient of x^3.

A -240

B -160 ✓

C 160

D -40

WORKED SOLUTION & VERIFICATION

Using the binomial theorem, the expansion of $(1+ax)^n$ is given by:

$$(1+ax)^n = 1 + \binom{n}{1}(ax) + \binom{n}{2}(ax)^2 + \binom{n}{3}(ax)^3 + \dots$$

1. **Equate the coefficient of x:**

$$\binom{n}{1}a = na = -12 \implies a = -\frac{12}{n}$$

2. **Equate the coefficient of x^2:**

$$\binom{n}{2}a^2 = \frac{n(n-1)}{2}a^2 = 60 \implies n(n-1)a^2 = 120$$

Substitute $a = -\frac{12}{n}$:

$$n(n-1)\left(-\frac{12}{n}\right)^2 = 120$$

$$n(n-1)\cdot\frac{144}{n^2} = 120 \implies \frac{n-1}{n} = \frac{120}{144} = \frac{5}{6}$$

Solving for n:

$$6(n-1) = 5n \implies 6n-6 = 5n \implies n = 6$$

Then, find a:

$$a = -\frac{12}{6} = -2$$

3. **Calculate the coefficient of x^3:**

$$\text{Coefficient of } x^3 = \binom{n}{3}a^3 = \binom{6}{3}(-2)^3 = 20 \times (-8) = -160$$

Common Traps:

- -160: Forgetting that cubing a negative number yields a negative result, i.e., $(-2)^3 = -8$, not $+8$.
- -40: Failing to raise a to the power of 3, computing $\binom{6}{3}a = 20(-2) = -40$.
- -240: Incorrectly calculating the combination $\binom{6}{3}$ or taking $\binom{6}{2}(-2)^3 = 15(-16) = -240$. [M1][A1]

13 MCQ

An arithmetic progression (h_u) has first term $h_1 = 24$ and common difference $d = 17$. Find the term h_5.

A 116	B 109
C 75	D **92** ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** Using $h_n = h_1 + (n-1)d$, we have $h_5 = 24 + (5-1)(17) = 92$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 92. **[A1]**

14 MCQ

The polynomial $p(x) = 2x^3 + ax^2 - 5x + 6$ is divisible by $x - 2$. Find the value of the constant a.

A -3 ✓	B -6
C 3	D 0

WORKED SOLUTION & VERIFICATION

By the Factor Theorem, if $p(x)$ is divisible by $x - 2$, then $p(2) = 0$.

Substitute $x = 2$ into the polynomial expression:

$$p(2) = 2(2)^3 + a(2)^2 - 5(2) + 6$$

$$p(2) = 2(8) + 4a - 10 + 6 = 16 + 4a - 10 + 6 = 4a + 12$$

Setting $p(2) = 0$:

$$4a + 12 = 0 \implies 4a = -12 \implies a = -3$$

Common misconceptions:

- Using the root $x = -2$ instead of $x = 2$ leads to $2(-8) + 4a - 5(-2) + 6 = 4a = 0 \implies a = 0$.
- A sign error when solving the linear equation $4a + 12 = 0$ incorrectly gives $a = 3$. [M1][A1]

15 MCQ

The polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 4)$. What is the value of the constant a?

A -10.00	B 7.00
C **-7.00** ✓	D -5.00

WORKED SOLUTION & VERIFICATION

By the Factor Theorem, $P(4) = 0 \implies 2(4)^2 + a(4) - 4 = 0 \implies 32 + 4a - 4 = 0 \implies a = -7.00$. [M1][A1]

16 MCQ

In this problem, the polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 5)$. What is the value of the constant a?

A 9.20	B **-9.20** ✓
C -12.20	D -7.20

WORKED SOLUTION & VERIFICATION

By the Factor Theorem, $P(5) = 0 \implies 2(5)^2 + a(5) - 4 = 0 \implies 50 + 5a - 4 = 0 \implies a = -9.20$. [M1][A1]

17 MCQ

Observe that the polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 6)$. What is the value of the constant a?

A **-11.33** ✓	B -9.33
C -14.33	D 11.33

WORKED SOLUTION & VERIFICATION

By the Factor Theorem, $P(6) = 0 \implies 2(6)^2 + a(6) - 4 = 0 \implies 72 + 6a - 4 = 0 \implies a = -11.33$. [M1][A1]

18 MCQ

Suppose the polynomial $P(x) = 2x^2 + ax - 4$ is exactly divisible by $(x - 7)$. What is the value of the constant a?

A 13.43	B **-13.43** ✓
C -11.43	D -16.43

WORKED SOLUTION & VERIFICATION

By the Factor Theorem, $P(7) = 0 \implies 2(7)^2 + a(7) - 4 = 0 \implies 98 + 7a - 4 = 0 \implies a = -13.43$. [M1][A1]

19 MCQ

Determine the solution to the exponential equation $2^{x+7} = 5^x$, giving x in terms of common logarithms.

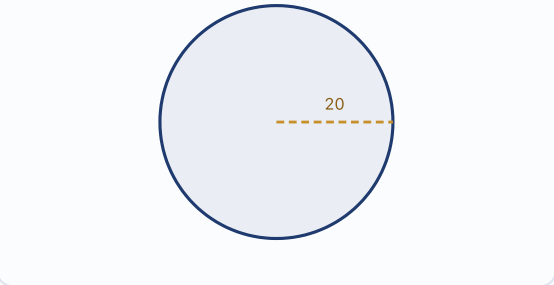
A $x = \frac{7 \ln 5}{\ln 5 - \ln 2}$	B $x = \frac{7 \ln 2}{\ln 5 - \ln 2}$ ✓
C $x = \frac{7}{\ln 5 - \ln 2}$	D $x = \frac{\ln 2}{7(\ln 5 - \ln 2)}$

WORKED SOLUTION & VERIFICATION

Take natural logs of both sides: $(x + 7)\ln 2 = x \ln 5 \implies x \ln 2 + 7 \ln 2 = x \ln 5 \implies x(\ln 5 - \ln 2) = 7 \ln 2 \implies x = \frac{7 \ln 2}{\ln 5 - \ln 2}$. [M1][A1]

20 MCQ

A sector of a circle of radius $r = 20$ cm has central angle $\theta = 1.5$ rad. What is the area of the sector?



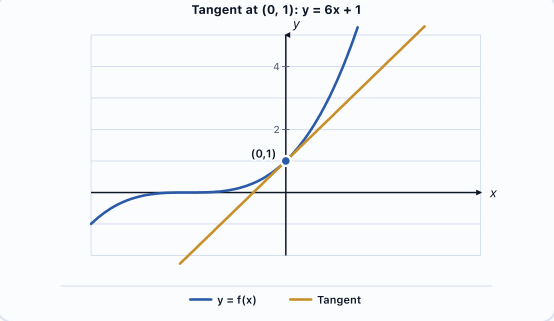
- **A** $300.0\ \text{cm}^2$ ✓
- **B** $30.0\ \text{cm}^2$
- **C** $600.0\ \text{cm}^2$
- **D** $305.0\ \text{cm}^2$

WORKED SOLUTION & VERIFICATION

Sector area formula: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(20)^2(1.5) = 300.0\ \text{cm}^2$. [M1][A1]

21 MCQ

Find the equation of the tangent line to the curve $y = (2x+1)^3$ at the point where $x = 0$.



- **A** $y = 6x + 1$ ✓
- **B** $y = 2x + 1$
- **C** $y = 6x - 1$
- **D** $y = 6x + 3$

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(2x+1)^2 \cdot (2) = 6(2x+1)^2$. At $x = 0$, slope $m = 6$. Equation: $y - 1 = 6(x - 0) \implies y = 6x + 1$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: $y = 6x + 1$. **[A1]**

22 MCQ

Determine the numerical value of the definite integral: $\int_0^1 (3x+1)^3\,dx$.

- **A** $\frac{255}{12}$ ✓
- **B** $\frac{255}{4}$
- **C** $\frac{255}{3}$
- **D** $\frac{259}{12}$

WORKED SOLUTION & VERIFICATION

Substitute $u = 3x+1 \implies du = 3\,dx$. Integral: $\frac{1}{3}\left[\frac{u^4}{4}\right]_1^4 = \frac{(4)^4-1^4}{12} = \frac{255}{12}$. [M1][A1]

23 MCQ

In this problem, the function $f(x) = \frac{4x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

- **A** $f^{-1}(x) = \frac{2x-1}{x-4}$
- **B** $f^{-1}(x) = \frac{2x+1}{x-4} \quad (x \neq 4)$ ✓
- **C** $f^{-1}(x) = \frac{x-2}{4x+1}$
- **D** $f^{-1}(x) = \frac{4x-2}{x+1}$

WORKED SOLUTION & VERIFICATION

Let $y = \frac{4x+1}{x-2} \implies y(x-2) = 4x+1 \implies yx-2y = 4x+1 \implies x(y-4) = 2y+1 \implies x = \frac{2y+1}{y-4}$. Swap variables to obtain $f^{-1}(x) = \frac{2x+1}{x-4}$. [M1][A1]

24 MCQ

In the expansion of $(1+\frac{x}{2})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?

- **A** 11
- **B** 8
- **C** 5
- **D** 9 ✓

WORKED SOLUTION & VERIFICATION

Coeff of x is $\frac{n}{2}$. Coeff of x^2 is $\frac{n(n-1)}{2(4)}$. Given $\frac{n(n-1)}{2\cdot 4} = 2\left(\frac{n}{2}\right) \implies \frac{n-1}{2\cdot 2} = 2 \implies n-1 = 4(2) = 8 \implies n = 9$. [M1][A1]

25 MCQ

Considering the expansion of $(1+\frac{x}{3})^n$, the coefficient of x^2 is twice the coefficient of x. Given $n > 0$, what is the value of n?

- **A** 12
- **B** 13 ✓
- **C** 7
- **D** 15

WORKED SOLUTION & VERIFICATION

Coeff of x is $\frac{n}{3}$. Coeff of x^2 is $\frac{n(n-1)}{2(9)}$. Given $\frac{n(n-1)}{2\cdot 9} = 2\left(\frac{n}{3}\right) \implies \frac{n-1}{2\cdot 3} = 2 \implies n-1 = 4(3) = 12 \implies n = 13$. [M1][A1]